

POLTAVKA, Russian Federation

WMO: 287860

Lat: **54.3561N**Lon: **71.7661E**Elev: **410**StdP: **14.48**Time Zone: **6.00 (E06)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.6	-23.7	-33.8	0.8	-28.4	-28.9	1.1	-23.5	20.5	6.7	18.8	10.6	3.3	180	0.611

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.3	88.1	65.3	84.5	64.6	81.3	63.7	69.0	81.0	67.4	79.0	66.0	77.3	7.7	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.2	94.6	73.6	63.5	88.9	72.1	61.8	83.6	70.9	33.4	81.2	32.1	78.7	31.1	77.5	76.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.2	17.2	15.4	-32.8	93.3	6.6	3.1	-37.6	95.6	-41.4	97.4	-45.1	99.1	-49.9	101.4		
(5)			-33.3	71.7	6.6	1.9	-38.0	73.0	-41.8	74.1	-45.5	75.2	-50.3	76.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	36.2	0.3	4.1	18.6	40.1	54.9	64.9	67.0	63.4	52.6	39.0	20.2	7.0
	DBStd	26.27	13.78	13.10	11.39	11.24	9.32	8.10	6.62	7.39	8.27	9.21	13.54	14.07
	HDD50	6841	1540	1286	973	331	57	4	0	2	64	356	895	1333
	HDD65	10837	2005	1706	1438	747	332	101	58	120	381	806	1345	1798
	CDD50	1797	0	0	0	34	208	452	527	419	142	15	0	0
	CDD65	318	0	0	0	0	18	99	120	72	9	0	0	0
	CDH74	3309	0	0	0	11	304	1045	1091	728	129	1	0	0
	CDH80	1008	0	0	0	1	71	358	320	226	32	0	0	0
Wind	WSAvg	7.2	7.3	7.4	8.2	8.3	8.0	6.4	5.8	5.8	6.6	7.4	7.8	7.9
Precipitation	PrecAvg	14.4	0.7	0.6	0.6	0.9	1.2	1.9	2.5	1.9	0.9	1.2	1.1	0.9
	PrecMax	22.4	1.5	1.2	1.6	2.4	3.5	4.6	4.6	3.7	2.4	3.1	3.0	1.7
	PrecMin	8.8	0.1	0.1	0.0	0.2	0.2	0.7	0.5	0.6	0.2	0.1	0.3	0.2
	PrecStd	3.1	0.4	0.3	0.4	0.6	0.7	0.9	1.1	0.9	0.6	0.7	0.6	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.9	33.4	44.8	74.7	86.5	92.0	93.0	91.1	84.3	68.8	45.5	34.7
		MCWB	31.8	31.8	39.4	54.8	59.3	65.4	67.9	65.6	59.6	52.5	42.1	32.6
	2%	DB	28.8	30.7	38.1	67.9	81.1	88.0	86.8	85.7	77.6	63.2	42.2	32.2
		MCWB	27.4	29.2	35.0	51.2	58.1	65.0	66.4	65.5	57.3	50.5	39.3	31.1
	5%	DB	25.2	27.4	35.7	63.2	77.2	84.1	83.1	81.5	71.7	58.0	39.2	29.6
		MCWB	24.0	26.2	33.4	49.9	57.1	64.2	66.2	64.3	56.1	47.7	36.5	28.4
	10%	DB	21.2	23.6	33.8	57.7	72.8	80.0	80.1	77.7	66.6	52.9	35.7	25.6
		MCWB	20.1	22.3	32.0	47.2	55.4	62.9	65.6	63.1	54.0	45.8	33.6	24.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	31.9	32.1	39.4	56.7	62.7	70.4	71.4	69.8	62.1	55.1	43.2	33.2
		MCDB	32.8	33.2	44.5	69.9	76.4	83.9	82.6	82.8	77.9	66.1	44.9	33.9
	2%	WB	27.7	29.3	35.4	53.3	60.2	68.0	69.6	67.6	59.5	52.0	39.6	31.1
		MCDB	28.8	30.7	37.5	64.6	76.6	81.3	81.1	79.5	72.7	59.8	42.0	32.1
	5%	WB	24.1	26.2	33.7	50.4	58.5	66.2	68.1	66.1	57.4	48.8	36.4	28.4
		MCDB	25.3	27.4	35.5	61.9	74.0	79.4	79.1	77.5	69.4	56.4	38.8	29.8
	10%	WB	20.1	22.3	32.0	47.6	56.7	64.3	66.7	64.4	55.1	46.0	33.8	24.2
		MCDB	21.1	23.5	33.9	57.1	70.1	77.0	77.3	75.4	64.9	52.3	35.6	25.4
Mean Daily Temperature Range	5% DB	MDBR	13.3	14.5	14.7	17.4	21.9	20.2	18.3	19.3	18.5	14.4	11.6	13.2
		MCDBR	14.7	13.5	12.5	26.5	28.7	26.0	23.5	26.3	26.8	23.9	11.8	13.5
		MCWBR	14.6	13.0	10.6	13.6	12.1	9.9	9.1	10.5	12.1	13.7	9.9	13.2
	5% WB	MCDBR	14.6	13.3	11.6	24.3	25.3	22.3	19.6	22.1	23.6	20.8	11.0	13.5
		MCWBR	14.5	12.8	10.4	13.8	12.6	9.6	8.7	10.3	11.6	13.0	9.9	13.3
Clear-Sky Solar Irradiance	taub	0.287	0.317	0.375	0.409	0.406	0.410	0.435	0.430	0.372	0.352	0.323	0.289	
	taud	2.049	2.018	1.951	2.131	2.220	2.295	2.212	2.190	2.314	2.253	2.117	2.021	
	Ebn at Noon	197	234	245	257	266	266	256	248	250	223	181	165	
	Edh at Noon	23	33	44	43	42	39	42	40	31	26	21	19	
All-Sky Solar Radiation	RadAvg	335	684	1170	1477	1811	1927	1772	1462	1043	539	264	217	
	RadStd	23	26	104	138	146	112	129	113	75	74	36	27	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (12 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page